RUDRAEDGE AEROSPACE

Project - Rudra Edge RE-T1(Spectra)

Daily Engineering Log

1. TEAM MEMBER INFORMATION

Engineer Name: Suraj Nadda
Designation / Role: Software & Data Engineer
Task Order Reference: 05 October 2026 – Data Flow Architecture

2. DAILY LOG ENTRY FORM

• **Date:** 05 October 2026

• **1. Assigned Task:** Data Architecture Study and Multisensor Bus Topography

---

• **2. What Was Studied:**

Studied sensor data flow from multiple input sources through communication, data acquisition, parsing, validation, synchronization, preprocessing, telemetry and storage/analysis. Reviewed the conceptual arrangement of multiple sensors on a shared communication architecture and the purpose of communication interfaces such as I²C, SPI and UART. Also studied how structured and validated data can be prepared for downstream analysis, monitoring and future processing.

• **3. What Was Implemented:**

Prepared a conceptual RE-T1 data-flow architecture and multi-sensor bus topography. Documented the flow from sensors through communication and acquisition to processing, validation and storage/analysis. Added clear interface-level notes for data collection, synchronization and validation so that the architecture can be extended later when exact sensor-to-bus hardware assignments are confirmed.

• **4. Tools / Libraries Used:**

VS Code for technical documentation and diagrams.net (draw.io) for the architecture diagram. No programming library was required for this documentation task; the work was focused on architecture mapping, data-flow understanding and technical documentation.

• **5. Output File Name : suraj_rudra**_task/Day1/notes/01_data_flow_architecture_notes

Path: https://github.com/surajnadda29/suraj_rudra_t1.git

• **6. Problems Encountered:**

None. No problem encountered during studying data flow Architecture

---

• **7. How Problem Was Resolved:**

No issue was encountered during the task ,so no correction action was required. The data-flow architecture and multi-sensor bus topography were completely as planned

---

• **8. Next Task Scheduled:**

6 Oct 2026 - Python Environment Setup Python environment organization and robust software dependency setup

---

• **9. Personal Expenditure: ₹0 (Verified Mandate)**

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, architecture documentation, data-flow notes, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastructure.

Candidate/Engineer Signature:  Date: 05/10/2026

Full Name: Suraj Nadda

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

[ ] Approved / Completed

[ ] Needs Revision / Delayed

[ ] Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____/____/2026